

Four problems — answers

1 — Classify by sign

The rule: sign of the direct path $X \rightarrow Z$, against the **product** of the signs along $X \rightarrow Y \rightarrow Z$. Same $\rightarrow$ coherent; opposite $\rightarrow$ incoherent.

Now you: $X \xrightarrow{-} Y$, $Y \xrightarrow{+} Z$, $X \xrightarrow{+} Z$.

Direct $+$; indirect $(-)(+) = -$. Opposite, so **incoherent**. With X repressing Y and both X and Y activating Z , this is **incoherent type 4** in Mangan & Alon's numbering — their own words, p. 11982: "in the type 4 incoherent FFL, when S_x turns on, Z is first induced by the joint action of X and Y . Meanwhile, Y production is repressed by X ." Their Table 2 records it as rare — 1 in *E. coli*, 0 in yeast — and gives the reason: with an AND gate its steady-state output responds to *neither* S_x nor S_y , so it has less function than types 1 and 2, which respond to both.

Note what it is good at, because it is the opposite of what the abundance suggests: Table 2 marks type 4 as the only **strong** pulser on an S_x on-step, where type 1 is marked *weak*. Rare in nature, useful in a design.

2 — The delay table

$$t_D = \ln \left[\frac{Y_{\max} - Y_{\min}}{Y_{\max} - K_{yz}} \right] \text{ with } Y_{\max} = 1, Y_{\min} = 0, \text{ so } t_D = \ln \left[\frac{1}{1 - K_{yz}} \right].$$

K_{yz}	FORMULA	SIMULATED AT $H = 2$	OFF STEP
0.2	$\ln 1.25 = 0.23$	0.35	0
0.5	$\ln 2 = 0.69$	0.75	0
0.8	$\ln 5 = 1.61$	1.02	0

The OFF-step delay is **0** in every row, and that is not an approximation: the measured value is **−0.0000** lifetimes. With an AND gate, removing X opens the gate immediately no matter what Y is doing, so the FFL and the simple circuit fall on exactly the same curve.

Why the formula fails at $K_{yz} = 0.8$. It **overestimates**, 1.61 against 1.02 — and notice it errs the *other* way at $K_{yz} = 0.2$ (0.23 against 0.35). The error is not a consistent offset, so it is not a fudge factor you can carry.

What it is: the formula is the **sharp-gate limit**. It assumes Z is strictly off until Y crosses K_{yz} , which is only true for $H \gg 1$. Raising H in the simulation drives every row onto the formula — at $H = 40$ all three agree to within **0.04** lifetimes — so the whole discrepancy is gate softness and nothing else.

At $H = 2$ the gate leaks. $f(Y, K_{yz})$ is a few percent open even when Y is well below K_{yz} , and because we matched the two circuits to the same steady state, the FFL's Z promoter is strong enough that a few percent is enough to start it climbing. That leak is what shortens the wait at large K_{yz} , where the formula is predicting a very long one.

The design reading, and it is the one worth carrying: **a threshold is only as good as it is sharp, and sharpness is cooperativity**. The parameter that opened the toggle's wedge on Thursday is the same parameter that decides whether a delay you designed on paper is the delay you get.

The OR gate. The delay moves to the **OFF step**, and the ON step becomes immediate. With OR, either input alone turns Z on, so an ON step of X fires Z at once; on an OFF step Z stays up until Y has decayed below its threshold. Mangan & Alon, Table 1: the FFL's functions survive the swap, with the sign sensitivity reversed.

3 — The adaptation error

Integrating the incoherent type-1 with $B_y = 0.4$, $K_{yz} = 0.5$, $\beta_z = 6$, stepping X from 0 to 1:

- **peak** = 1.099, reached at $t = 1.07$ lifetimes
- **final** = 0.681
- **adaptation error** = $0.681/1.099 = 0.62$

The two instruments. An adaptation error near 0 is a circuit that returns almost to where it started even though the input is still on: it reports *change* and is blind to level. That is what bacterial chemotaxis does, and it is why a cell swimming up a gradient responds to getting better rather than to being in a good place. An error near 1 barely pulsed at all: it reports *level*, and the incoherent wiring has bought only a faster approach to it. Both are useful; you have to say which one you are building, because the same three genes give you either depending on B_y and the strength of the repression.

4 — The design problem

Two answers are defensible, and the difference between them is the whole item. Both are coherent, both take an AND gate, and Mangan & Alon's Table 1 — projected at minute 28 — records both as delaying the S_x on-step and neither as delaying the off-step. The incoherent types are out: they respond immediately and then adapt.

Answer A: coherent type 1 with an AND gate. The delay you derived in item 2 is a persistence filter: the output fires only if the input stays on longer than t_D , because a shorter pulse never lets Y rise to K_{yz} . Lifetimes are near 30 minutes, so one hour is $t_D \approx 2$ lifetimes. Setting $\ln[1/(1 - K_{yz})] = 2$ gives

$$K_{yz} = 1 - e^{-2} = 0.865,$$

i.e. put Y 's operator at about 87% of Y 's maximum level — a weak-affinity site on the $Y \rightarrow Z$ arm.

And answer A does not reach an hour, which is half the point of the item.

Item 2 already warned that the formula overestimates in this regime. Simulate $K_{yz} = 0.865$ at $H = 2$ and the delay is 1.06 lifetimes, about half an hour, not one. Push the threshold higher and it barely moves: **0.90 $\rightarrow$ 1.08, 0.95 $\rightarrow$ 1.11, 0.99 $\rightarrow$ 1.13. The delay saturates near 1.1 lifetimes, and no value of K_{yz} reaches 2.** You cannot buy an hour by moving the operator, because the leak through a soft gate sets a ceiling the threshold cannot raise.

The obvious second knob fails too, but not for the reason you would guess. Slowing Y looks like it should stretch the delay, since α_y^{-1} sits in front of the formula — but α_y also sets the ceiling, $Y_{\max} = \beta_y/\alpha_y$, so a longer-lived Y climbs past K_{yz} sooner relative to its own scale. The two effects nearly cancel. At $K_{yz} = 0.865$ the measured delay goes **1.06 $\rightarrow$ 1.23 $\rightarrow$ 1.25** for $\alpha_y = 1, 0.5, 0.25$ — a 17% gain, not the doubling the prefactor promises. The sharp-gate formula, with $Y_{\max}$ tracking α_y , actually predicts the delay **falls: 2.07 $\rightarrow$ 1.15 $\rightarrow$ 0.99.** Slow Y down and you must raise K_{yz} in step, and then you are back at the same ceiling.

What works is sharpening the gate. Holding $K_{yz} = 0.8$ and raising the cooperativity of the $Y \rightarrow Z$ interaction: $H = 2 \rightarrow 1.02$, $H = 4 \rightarrow 1.41$, $H = 6 \rightarrow 1.60$, $H = 10 \rightarrow 1.73$. To specify an hour you need a cooperative **activator** at Y 's site on the Z promoter — a dimer, or a pair of binding sites. Note that none of the numbers above reaches 2 on its own: the combination that does is $H = 10$ with $K_{yz} = 0.865$, giving $t_D = 2.01$ lifetimes. By then the formula is telling the truth, so you can place K_{yz} from it rather than by search.

Answer B: coherent type 4 with an AND gate — and this one hits the hour at $H = 2$. You have already classified this wiring: it is item 1(c). $X \dashv Y$, $Y \dashv Z$, $X \rightarrow Z$. Now the delay is built on the *other* side of the threshold. When X turns on, Y stops being made and *decays* past K_{yz} , and only then is Z released. So

$$t_D = \alpha_y^{-1} \ln \left[\frac{Y_{\max}}{K_{yz}} \right],$$

which runs to infinity as $K_{yz} \rightarrow 0$ instead of hitting a wall at $Y_{\max}$. There is no ceiling to saturate against, and the soft gate no longer fights you — it is working on a decaying exponential, not on the approach to a plateau. Simulated at $H = 2$ with the same parts:

K_{yz}	0.5	0.2	0.168	0.1	0.05
t_D (on)	0.91	1.82	2.00	2.56	3.35
t_D (off)	0	0	0	0	0

$K_{yz} \approx 0.17$ gives exactly one hour at 30-minute lifetimes, with the shut-off still immediate. No cooperative activator required.

The design reading, and it is the point of the item. Both circuits delay. What separates them is *which side of the threshold the delay is built on*:

- Type 1 waits for Y to **climb** to K_{yz} . The climb is bounded by $Y_{\max}$, so the threshold you can usefully set is bounded too, and a soft gate leaks before you get there. **Cooperativity is a prerequisite.**
- Type 4 waits for Y to **fall** past K_{yz} . A decaying exponential has no floor, so the threshold can go as low as you can build it. **Cooperativity is a luxury.**

That is a sharper statement than "you need a cooperative repressor," and it generalizes: whenever a timer is built out of a concentration crossing a threshold, ask whether it is approaching a ceiling or leaving one. Thursday's engineerability conclusion still stands for type 1 — the parameter the design most depends on is the one you buy by choosing a protein — but the right first move is to check whether you have to be in that regime at all.

Marking: full credit for either answer, argued. A student who reaches type 4 and sees why it escapes the ceiling has done something this session did not do from the front, and should be told so. A student who answers type 1 and reports the ceiling honestly has done exactly what item 2 trained them to do.

Why you rarely see type 4 in the wild. Mangan & Alon count 1 in *E. coli* and 0 in yeast, against 28 and 26 for type 1. Their argument is evolutionary, not engineering (p. 11984): types 3 and 4 need X to act with

opposite signs on Y and Z , which is "well established" as biologically feasible but leaves the circuit with less steady-state function. Rare is not the same as unbuildable, and this is a course about building.

One part of the specification is free in both answers: with an AND gate the shut-off is immediate, which is what was asked. An OR gate would have given you a circuit that fires instantly and lingers — the exact opposite, from the same three genes.